# Improving Lane Detection Performance Using Morphological Operations and Adaptive Binarization

Gyuwon Na[1], Junewoo Choi[2], Deokwoo Lee[2,*]

[1]Department of DigiPen Game Engineering, Keimyung University

[2]Department of Computer Engineering, Keimyung University

uoyuvvou@gmail.com, jwoochoi99@gmail.com, dwoolee@kmu.ac.kr

**Abstract—** This paper proposes a lane-detection method that combines morphological closing with Otsu binarization to improve the performance of geometry-based lane-recognition algorithms. Experiments on the TuSimple dataset show that the proposed method produces more stable binary masks than conventional fixed-threshold approaches under abrupt illumination changes. It also improves noise suppression and tracking consistency while preserving real-time processing efficiency.

## 1. Introduction

Lane detection is a fundamental component of autonomous-driving and advanced driver-assistance systems (ADAS), enabling a vehicle to maintain a safe trajectory. Accurate lane perception supports lane-departure warnings, lane-keeping assistance, and automated steering control, and is therefore directly related to system reliability.

Conventional lane-detection algorithms are commonly implemented as sequential pipelines that include color-space conversion, edge detection, the Hough transform, and sliding-window-based lane tracking. However, these approaches are vulnerable to the variability of real road environments. Their performance can deteriorate significantly under abrupt illumination changes and complex conditions such as tunnel entrances, shadows, nighttime driving, and damaged lane markings.

This paper proposes a preprocessing method that combines morphological closing, which strengthens the continuity of lane-candidate regions, with Otsu thresholding, which automatically adapts to illumination changes. The resulting binary masks improve the reliability of subsequent lane detection across diverse driving conditions.

## 2. Related Work

Traditional lane-detection pipelines generally consist of color-space conversion, Canny edge detection, the Hough transform, and lane fitting. Considerable research has focused on reducing noise and restoring discontinuous lane markings within these pipelines.

Tang and Luo applied morphological operations after color-feature extraction to improve broken lane markings and suppress noise [1]. Their experiments showed that morphological closing was effective in maintaining lane continuity and removing noise, although performance degraded in low-light environments and the method had limitations for real-time processing. Kim and Song also demonstrated that applying dilation after HSV conversion could reduce internal gaps in lane masks under real driving conditions [2].

Abrupt brightness changes caused by tunnel entrances, structural shadows, or tree cover make lane detection particularly difficult. To address this issue, adaptive thresholding and Otsu-based methods have been investigated. Huang et al. proposed a multi-threshold selection strategy for LiDAR-based lane detection that adapts threshold intervals to reflection intensity [3]. Although the method improved robustness to brightness variation, performance still degraded on dashed lanes because discontinuous intervals were difficult to detect reliably.

## 3. Problem Statement

Geometry-based lane-detection algorithms exhibit two major limitations in real-world environments.

First, HSV-based extraction is sensitive to the selected color range. A narrow range suppresses noise but can fragment the interior of a lane marking, producing incomplete detection. More fundamentally, HSV segmentation becomes unreliable when the background and lane markings have similar color distributions. Severely worn or faded lanes, as well as dirt, dust, and leaves on the road, blur the distinction between lane pixels and the surrounding pavement. Under such conditions, precise HSV thresholds are difficult to define and portions of the lane may be lost [4].

Second, conventional methods are highly sensitive to illumination changes. Brightness conditions can change rapidly at tunnel entrances and exits, beneath bridges, and under tree shadows. Fixed thresholds cannot immediately adapt, making it difficult to distinguish lane boundaries from the background consistently [5]. This instability can delay lane recognition and adversely affect driving safety. Excessive exposure from headlights at night or deep shadows cast by roadside objects can likewise obscure the boundary between lane markings and the road surface, causing detection failure [6].

Therefore, the combination of fixed color ranges and sensitivity to illumination makes stable lane detection difficult across varied environments.

## 4. Proposed Method

### 4.1 Morphological Closing

Morphological operations modify object shapes according to image structure and are widely used to remove noise or emphasize specific geometric patterns. Closing is a compound operator that applies dilation followed by erosion:

$$A \bullet B = (A \oplus B) \ominus B, \tag{1}$$

where $A$ is the input binary image and $B$ is the structuring element.

Dilation expands object boundaries and fills small holes or disconnected regions. The subsequent erosion removes isolated noise and restores the object boundary, preserving a stable overall shape. In an HSV-based masking pipeline, closing can reconnect discontinuities caused by strict color thresholds. We therefore apply closing to strengthen the morphological continuity of lane candidates and obtain a more stable lane mask.

### 4.2 Otsu Adaptive Binarization

Otsu's method is a global thresholding technique that automatically determines an optimal threshold from an image histogram. Assuming that pixel intensities form two classes, it selects the threshold that maximizes between-class variance, equivalently minimizing the sum of within-class variances.

Because the threshold is recalculated independently for every frame, Otsu binarization can respond to dynamically changing illumination, such as the transition into or out of a tunnel. This compensates for the main weakness of fixed HSV ranges and enables the lane-detection pipeline to adapt more flexibly to environmental changes.

### 4.3 Implementation

The proposed algorithm was implemented in Python using OpenCV. Morphological closing was performed with a $5 \times 5$ rectangular structuring element, followed by Otsu binarization to generate an illumination-robust binary image. To reduce computational overhead, the Canny edge-detection stage was omitted. The processed binary mask was instead passed directly to the Hough-transform stage to detect candidate lane lines.

## 5. Experimental Results

The proposed method was evaluated on the TuSimple dataset under several illumination conditions. The *Conventional Method* applies HSV masking with a fixed threshold. *Closing Only* adds morphological closing to the conventional pipeline. *Closing* → *Otsu* is the proposed sequence, in which closing is followed by Otsu binarization. *Otsu* → *Closing* reverses the order to evaluate the influence of operation sequence. In each visualization, the red line indicates the image centerline and the blue line indicates the steering direction calculated from the detected lanes.

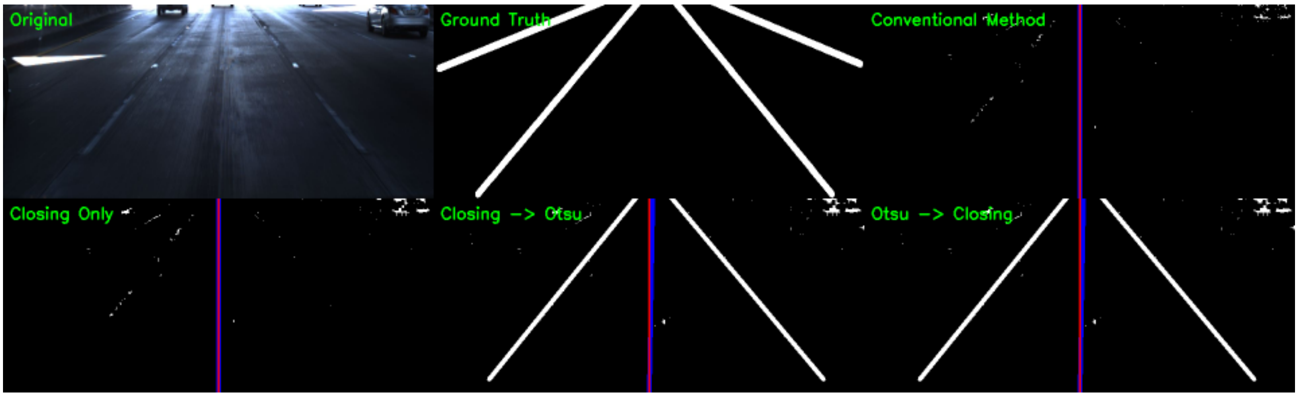


**Figure 1.** Comparison of lane-detection results under low-light conditions.

Figure 1 compares the methods in a low-light scene created by the abrupt shadow beneath a bridge or similar structure. The conventional pipeline and the closing-only variant failed completely because they could not distinguish lane pixels in the dark region. In contrast, the two pipelines that included adaptive thresholding produced meaningful lane detections. The proposed method achieved the same mean centerline pixel-distance error as the reversed pipeline, 4.84 px, while processing each frame approximately 1 ms faster. Thus, the proposed sequence preserved accuracy while improving real-time efficiency.

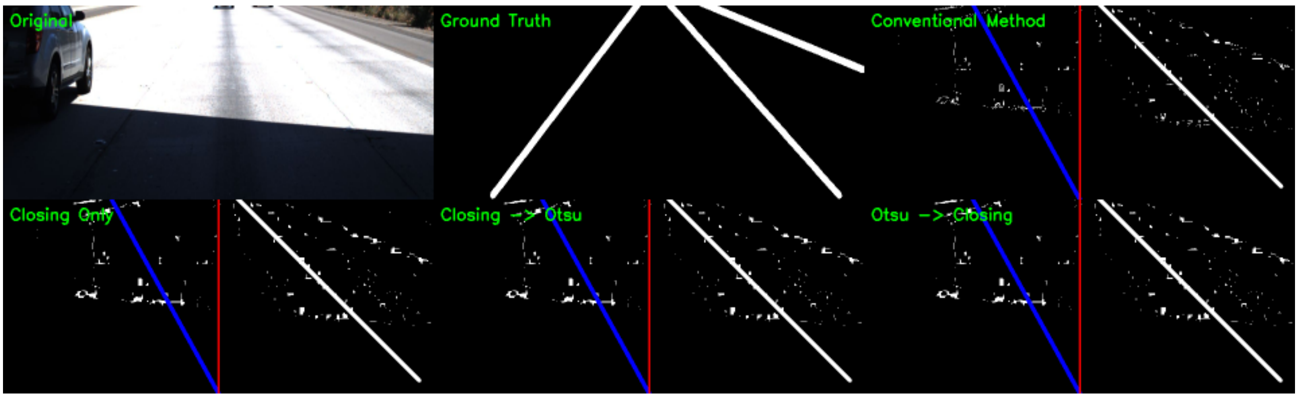


**Figure 2.** Comparison of lane-detection results under overexposed illumination.

Figure 2 shows a scene in which the road becomes strongly overexposed after leaving a dark region, significantly reducing contrast between the lane markings and pavement. Although most pipelines visually detected the lane, their detailed metrics differed. The proposed *Closing* → *Otsu* sequence produced the fewest noise pixels outside the lane region. Its mean centerline pixel-distance error was 13.18 px, slightly lower than the 13.2 px obtained by the alternative Otsu-based sequence. It also required less average processing time per frame.

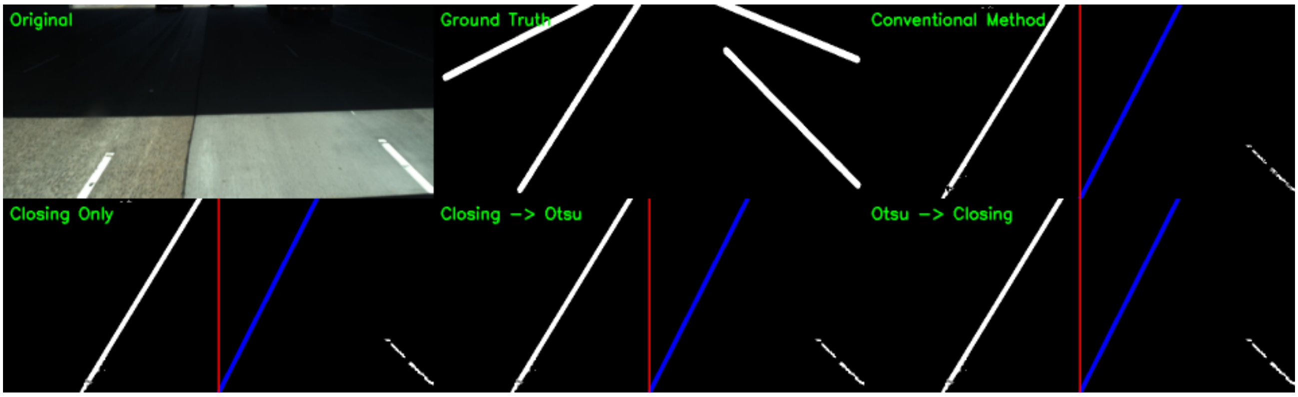


**Figure 3.** Comparison when structural shadows partially obscure the lane markings.

Figure 3 evaluates a case in which part of the lane is obscured by shadows from surrounding structures. All pipelines produced visually similar results, but the noise-penalty metric, which measures how far false-positive pixels lie from the true lane, revealed a substantial difference. The conventional method yielded a penalty of 73.7 px, whereas both *Closing* → *Otsu* and *Otsu* → *Closing* reduced the value to 9.1 px. The total number of noise pixels also decreased by approximately 30%, indicating that adaptive binarization produced a substantially cleaner mask and constrained false detections to regions close to the actual lane.

## 6. Conclusion

This paper analyzed the limitations of geometry-based lane-recognition algorithms and implemented a preprocessing method to improve their robustness. Morphological closing reconnects gaps within incompletely detected lane markings, while Otsu's method dynamically determines an optimal threshold for every frame. The combination enables more stable lane detection under abrupt illumination changes and partially damaged lane markings.

Nevertheless, the proposed approach remains primarily dependent on color information in the HSV space. Detection performance is therefore limited when lane markings have weak visual characteristics, such as heavily worn or damaged paint or lanes covered by dirt, dust, leaves, or other debris. Under such conditions, a color-based approach alone cannot guarantee reliable detection.

Future work will investigate multisensor fusion by combining camera images with LiDAR data. The objective is to improve lane-perception stability in extreme environments and to validate real-time performance and reliability across a broader range of real-road conditions.

## Acknowledgment

This work was supported by the Convergence and Open Sharing System project funded by the Ministry of Education and the National Research Foundation of Korea, and by the Basic Science Research Program through the National Research Foundation of Korea, funded by the Ministry of Education (No. 2022R1I1A3069352).

## References

[1] Z. Y. Tang and S. Y. Luo, "Combining Color Feature and Morphology Operations for Lane Detection," *International Journal of Research in Engineering and Science*, vol. 4, no. 8, pp. 81–85, Aug. 2016.

[2] K. B. Kim and D. H. Song, "Real Time Road Lane Detection with RANSAC and HSV Color Transformation," *Journal of Information and Communication Convergence Engineering*, vol. 15, no. 3, pp. 187–192, Sep. 2017.

[3] J. Huang, S. Yin, and L. Zhu, "An Efficient Multi-threshold Selection Method for Lane Detection Based on LiDAR," in *Proc. 2021 IEEE 4th International Conference on Electronics Technology*, Chengdu, China, 2021, pp. 1203–1207.

[4] A. Ghaida, H. Hikmarika, S. Dwijayanti, and B. Y. Suprapto, "Road and Vehicles Detection System Using HSV Color Space for Autonomous Vehicle," *Jurnal Ilmiah Teknik Elektro Komputer dan Informatika*, vol. 6, no. 1, pp. 42–53, Jun. 2020.

[5] Q. Huang and J. Liu, "Practical Limitations of Lane Detection Algorithm Based on Hough Transform in Challenging Scenarios," *International Journal of Advanced Robotic Systems*, vol. 18, no. 4, pp. 1–13, 2021.

[6] T. M. Hoang, N. R. Baek, S. W. Cho, K. W. Kim, and K. R. Park, "Road Lane Detection Robust to Shadows Based on a Fuzzy System Using a Visible Light Camera Sensor," *Sensors*, vol. 17, no. 11, p. 2475, 2017.